Lab-8

Aim: Image flashing in Raspberry Pi and Setting PC for interfacing with Raspberry Pi.

Appratus/Software Required: QNX Monentics IDE, VMware Workstation, Computer terminal, Raspberry Pi, 8GB SD Memory card, card reader

Image flashing Steps:

Step-1: Save QnX OS Image file rpi.img in hard disk.
Step-2: Search https://etcher.balena.io
Step-3: Download Etcher

Step-4: Select the below mentioned option and click "Download"

DOWNLOAD

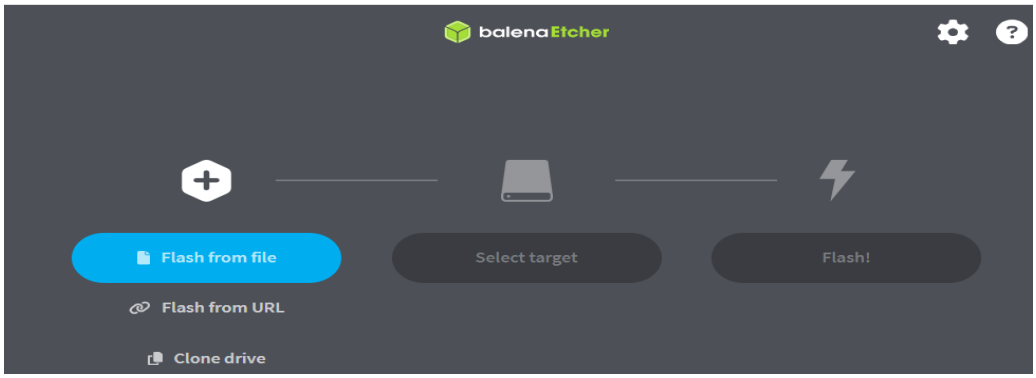
Download Etcher

ASSET	OS	ARCH	
ETCHER FOR WINDOWS (X86|X64) (INSTALLER)	WINDOWS	X86|X64	Download

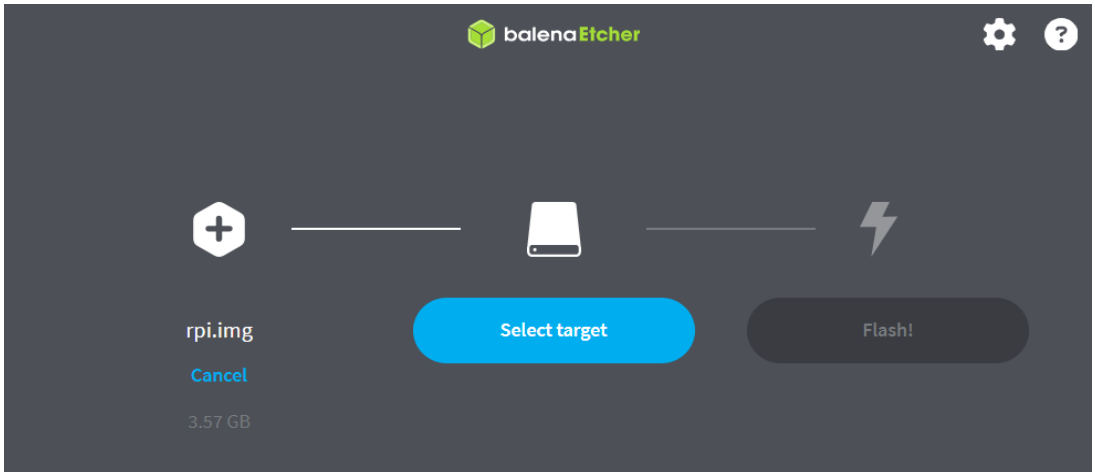
Step-5: Insert 8GB SD Memory card in the card reader and connect it to the USB connector of the PC.

Step-6: Open the "balenaEtcher" from the desktop.

Step-7: Click "Flash from file" and give the path of QnX OS Image file rpi.img



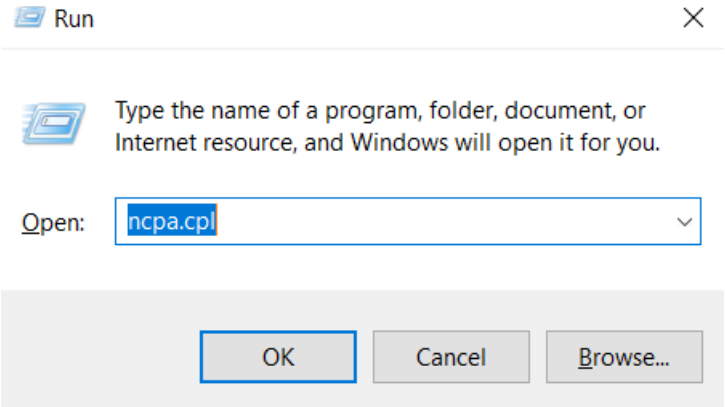
Step-8: Select the Target and give path of 8GM Memory card and then click “Flash”. Now the QnX OS Image file rpi.img has been flashed in the memory card.



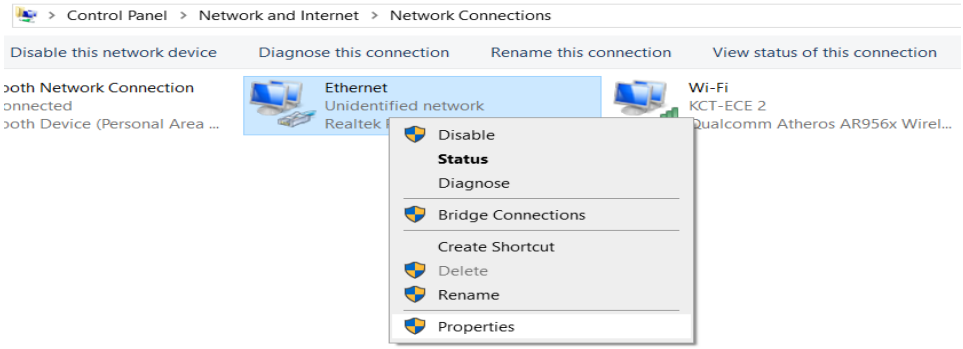
Step-9: Insert the Memory card in Raspberry Pi and connect it to PC

Step-10: The IP address of Raspberry Pi is programmed as 10.0.0.1 in QnX OS Image file rpi.img.

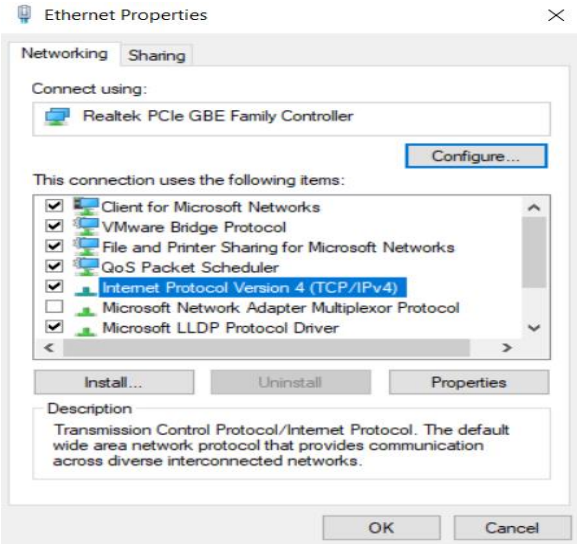
Step-11: PC IS Host and it’s IP address should be set as 10.0.0.2.
To Set IP of PC:
(I) Windows Symbol (in Keyboard) + R then “Run” window will pop-up and type “ncpa.cpl” in “Run” window “Open” field



(II) Right Click “Ethernet” and select “Properties”



(III) Double click “Internet Protocol Version 4 (TCP/IPv4)”



(IV) Set IP Address: 10.0.0.2

Set Subnet Mask: 255.255.255.0 and Click “OK"

Step-12: To Check Raspberry Pi is properly connected:

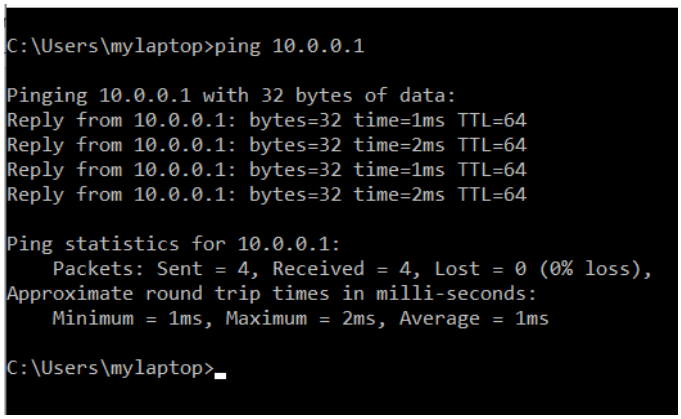
(I) Open Command Prompt

(II) Type ping 10.0.0.1 (which is IP address of Raspberry Pi)

If get following display then Raspberry Pi is connected properly

Ping time: around 1 or 2ms (approximately)

TTL: 64



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C:\Users\mylaptop>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: bytes=32 time=1ms TTL=64
Reply from 10.0.0.1: bytes=32 time=2ms TTL=64
Reply from 10.0.0.1: bytes=32 time=1ms TTL=64
Reply from 10.0.0.1: bytes=32 time=2ms TTL=64

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Users\mylaptop>
```